

EC400: Probability and Statistical Inference

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1 Introduction

Welcome to the Probability and Statistical Inference portion of EC400. This course is designed for incoming graduate students to prepare them for courses they will take during their degree, especially in econometrics. You can see a breakdown of the content of each of the five lectures below. You will probably recognize much of the material, and many of the ideas may be familiar from your previous studies. However, while many students are comfortable with the mathematics behind these ideas, they do not always have a fully intuitive understanding of the concepts or how to apply them. Achieving this level of fluency is the ultimate aim of the course.

There is also a pre-arrival quiz, which you can see on the moodle page. It is *not* necessary to have completed this before the lectures start. The main point of the quiz is to help you self-diagnose how well you understand the topics we are about to cover, so please use it to help identify your own gaps in understanding. If you wish to do some reading before term begins, then some optional textbook suggestions are given below. **You do not need to buy any textbooks for this course.** We will cover all of the material necessary for the EC400 PSI exam during the classes and lectures themselves.

The main prerequisite for your econometrics courses that we don't have enough time to talk about properly is linear algebra. The level required for this in econometrics courses is typically not too high, but it is a good idea to get yourself re-familiarized with things, especially if you are not coming straight from another degree. There are a couple of resources available:

- On the LSE website, you can also see a pdf of notes on *Linear Algebra for Econometrics*, which I have made. It contains some intuitive discussion of many of the basic ideas in

Linear algebra, and may well be helpful to look at if it's been a while since you've seen it.

- If you prefer lectures to written notes, this link contains a set of 18 short videos that I recorded a few years ago for another course at LSE, EC2C1. The videos cover very similar material to the above notes. Please see the *.readme* file in the folder for details on the content of each video.

Like the quiz, please note that you do not have to look at this before coming to LSE. The PSI exam will not test you on any of this prerequisite material. However, it is a good idea to take a look, as it may help you better comprehend both EC400 and the main courses.

Please do reach out if you have any questions in the meantime. Looking forward to seeing you in September!

- Tom

2 Content Overview

- **Lecture 1: Foundations.** Population targets and samples; randomized experiments, potential outcomes, average treatment effects and selection bias; identification vs. estimation. Sample statistics and quantiles; probability rules and frequentist and Bayesian interpretations. Parameters, estimators and estimates; repeated sampling and sampling distributions. Unbiasedness, bias, variance, efficiency, consistency and mean squared error; finite-sample vs. large-sample properties.

– Attempt Questions 1–5 from the pre-arrival quiz for self-assessment.

- **Lecture 2: Statistics I.** Probability rules, the union bound, random variables, cdfs and densities; expectation and linearity, indicator variables, variance, quantiles, skewness and kurtosis. Uniform and normal distributions; transformations. Joint, marginal and conditional distributions; Bayes' rule and odds; conditional moments, iterated expectations and total variance. Independence, conditional independence, iid observations, mean independence and zero covariance; links to OLS.

Appendix (non-examinable): defining expectation with integrals; proof of total variance. Characteristic functions and the substitution proof of LOTUS are also non-examinable.

– Attempt Questions 6–10 from the pre-arrival quiz for self-assessment.

- **Lecture 3: Statistics II.** Common discrete and continuous distributions and their applications; order statistics, minima and maxima. Covariance matrices and normal vectors; linear transformations; χ^2 , Student t and F distributions. Finite moments, tail behaviour, the Cauchy distribution and Jensen's inequality. Scalar, vector and matrix regression notation; matrix dimensions and products; OLS with and without an intercept; error covariance patterns and the conditional variance of OLS.

Appendix (non-examinable): deriving OLS by minimising squared residuals. Memorising the formulas in the opening distribution gallery is also non-examinable.

- Attempt Questions 11–15 from the pre-arrival quiz for self-assessment.

- **Lecture 4: Hypothesis Testing.** Null and alternative hypotheses; one-sided and two-sided tests; critical values, p -values and confidence intervals, including repeated-sampling coverage. Pivotal statistics, Z and T tests; exact and approximate inference; testing linear combinations of coefficients; conventional, White, cluster-robust and HAC standard errors. Type I and Type II errors, significance and power; Monte Carlo experiments; sample-size planning and minimum detectable effects. Multiple testing and Bonferroni; a signpost to Wald, likelihood-ratio and score/LM tests.

Appendix (non-examinable): further test-inversion examples, more general rejection regions and pre-testing bias. The Stata/R output examples and one-sided confidence-interval puzzle are also non-examinable.

- Attempt Questions 16–20 from the pre-arrival quiz for self-assessment.

- **Lecture 5: OLS and Asymptotics.** OLS in cross-sections and time series; assumptions A1–A4 and A3'; unbiasedness, efficiency and consistency; proof of unbiasedness. Convergence in probability and mean square. The law of large numbers, its sampling and moment conditions, and examples where averaging fails; stationarity and weak dependence. Matrix block identities and the proof of OLS consistency. Convergence in distribution, the central limit theorem and its conditions; proof of OLS asymptotic normality; approximate inference, consistency from asymptotic normality and the $n^{-1/2}$ convergence rate; course summary.

Appendix (non-examinable): continuous mapping and Slutsky's theorem; the scalar delta method; stochastic order notation; vector convergence and the Cramér–Wold device. Technical limit rules used in the main proofs are developed further in later courses.

- Attempt Questions 21–24 from the pre-arrival quiz for self-assessment. Question 25 covers the lecture's optional, non-examinable extension.

3 Suggested Further Reading

As mentioned above, all of the books below are **completely optional**: you do not need to buy any textbooks for this course. The lecture and class materials are self-contained for the EC400 PSI exam. These suggestions are simply alternative explanations and useful references if you would like to read around a topic. The *course collection* area of the LSE library contains several copies of many of these books.

- Larsen, R. and M. Marx, *An Introduction to Mathematical Statistics and Its Applications*, 5th (or 4th) edition, Pearson. Useful for distributions, moments, sampling distributions, estimation and testing.
- Greene, W.H., *Econometric Analysis*, 7th (or an earlier recent) edition, Appendices B–D. These appendices cover probability and statistics, estimation and inference, and large-sample distribution theory.
- Stock, J.H. and M.W. Watson, *Introduction to Econometrics*, 3rd or later edition. The probability and statistics review, regression notation and large-sample appendices are especially useful.
- Wooldridge, J.M., *Introductory Econometrics: A Modern Approach*, 5th or later edition, especially the introductory chapters and Appendices A–D:
 - Appendix A: Basic Mathematical Tools.
 - Appendix B: Fundamentals of Probability.
 - Appendix C: Fundamentals of Mathematical Statistics.
 - Appendix D: Summary of Matrix Algebra.
- Angrist, J.D. and J.-S. Pischke, *Mastering 'Metrics*, Princeton University Press. The opening chapters provide accessible intuition for targets, counterfactuals, randomized experiments and identification.
- Angrist, J.D. and J.-S. Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton University Press. This is a more advanced alternative to *Mastering 'Metrics*, with a deeper treatment of identification and the main empirical strategies used in modern applied econometrics.